Glossary: 03LagrangianMechanics

English	中文	Brief explanation
Physical space	物理空间	The ordinary three-dimensional Euclidean space $\mathbb{R}^3$ in which particles move.
Configuration	构型	The complete specification of the position of every particle in a system.
Configuration space	构型空间/位形空间	The $3N$-dimensional space of all possible configurations for a system of N particles; a point $q \in \mathbb{R}^{3N}$ encodes all positions.
Generalized coordinates	广义坐标	Any set of independent parameters $q_1, \dots, q_n$ that completely specifies the configuration of a system; they need not be Cartesian.
Generalized velocity	广义速度	The time derivative $\dot{q}_i$ of a generalized coordinate, giving the rate of change along the q_i direction in configuration space.
Phase space	相空间	The $6N$-dimensional space combining positions and momenta, $(q, p) \in \mathbb{R}^{6N}$; a point specifies the complete mechanical state in Hamiltonian mechanics.
Hamilton's principle	哈密顿原理 / 作用量取驻值原理	The actual path a system follows between two fixed configurations makes the action functional stationary: $\delta S = 0$.
Principle of least action	最小作用量原理	An informal name for Hamilton's principle; the stationary action is often a local minimum over short times.
Action	作用量	The functional $S[q] = \int_{t_i}^{t_f} L(q, \dot{q}, t)\, dt$ that assigns a number to each path in configuration space.
Lagrangian	拉格朗日量	The function $L(q, \dot{q}, t)$ whose integral gives the action; for many mechanical systems $L = T - V$.
Functional	泛函	A rule that maps an entire function (a path $q(t)$) to a single real number, e.g. the action $S[q]$.
Discrete approximation	离散近似	Approximating the action integral by a Riemann sum $\sum_k L_k \Delta t$ to clarify how S is evaluated.
Variation	变分	A small change in the input path, $q_\alpha(t) = q(t) + \alpha\eta(t)$, with fixed endpoints $\eta(t_i) = \eta(t_f) = 0$.
Euler–Lagrange equation	欧拉–拉格朗日方程	The necessary condition for stationary action: $\frac{d}{dt}\left(\partial L/\partial \dot{q}_k\right) - \partial L/\partial q_k = 0$, one per coordinate.

Fixed but arbitrary endpoints	固定但任意的端点	Boundary conditions $q(t_i)$ and $q(t_f)$ are held fixed during variation, but their values are not pre-determined; this yields a local equation valid for any pair.
Local differential equation	局部微分方程	The Euler–Lagrange equation holds at each instant along the path; once derived, the final endpoint need not be known in advance.
Initial-value problem	初值问题	Specifying $q(t_i)$ and $\dot{q}(t_i)$; standard ODE existence and uniqueness theorems guarantee a unique local solution.
Boundary-value problem	边值问题	Specifying $q(t_i)$ and $q(t_f)$; stationary paths may be unique, multiple, or nonexistent over long intervals.
Kinetic energy	动能	$T = \frac{1}{2}m\dot{\mathbf{r}}^2$ in Cartesian coordinates; in generalized coordinates it may become a quadratic form $T = \frac{1}{2}\sum_{i,j} A_{ij}(q)\dot{q}_i\dot{q}_j$.
Potential energy	势能	The function $V(q)$ from which conservative forces are derived; appears with a minus sign in the Lagrangian, $L = T - V$.
Proper time	固有时间	The Lorentz-invariant time measured along a particle's worldline in relativity; the relativistic free-particle action is proportional to it.
Worldline	世界线	The trajectory of a particle through spacetime.
Form invariance	形式不变性 / 协变性	The Euler–Lagrange equations retain exactly the same form in any choice of generalized coordinates.
Polar coordinates	极坐标	Coordinates (r, ϕ) in a plane; illustrate how the same Euler–Lagrange template yields equations that look more complicated but have identical structure.
Angular momentum	角动量	In polar coordinates $p_\phi = mr^2\dot{\phi}$; conserved when ϕ is cyclic.
Torque	力矩	The generalized force conjugate to an angular coordinate, $\tau = -\partial V/\partial\phi = rF_\phi$.
Constrained system	约束系统	A system whose motion is restricted by geometric or kinematic conditions, reducing the number of independent coordinates.
Degrees of freedom	自由度	The number of coordinates that can be independently varied in a small displacement.
Holonomic constraint	完整约束	A constraint expressible as an equation $f(\mathbf{r}_1, \ldots, \mathbf{r}_N, t) = 0$ relating coordinates (and possibly time).
Scleronomous constraint	定常约束	A holonomic constraint with no explicit time dependence.
Rheonomous	非定常约	

constraint	束	A holonomic constraint that depends explicitly on time.
Nonholonomic constraint	非完整约束	A constraint that cannot be written as a coordinate equation, e.g. an inequality or a nonintegrable velocity relation such as rolling without slipping.
Rolling without slipping	无滑动滚动	A velocity constraint relating translation to rotation; it is nonholonomic because it cannot be integrated into a pure coordinate relation.
Natural coordinates	自然坐标	Generalized coordinates in which the Cartesian-to-generalized transformation does not involve time explicitly; the kinetic energy is then a homogeneous quadratic function of the velocities.
Homogeneous quadratic function	齐次二次函数	A function f satisfying $f(\lambda x) = \lambda^k f(x)$ with $k = 2$; the kinetic energy in natural coordinates has this form.
Jacobian matrix	雅可比矩阵	The matrix $J^A{}_i = \partial x^A / \partial q_i$ of a coordinate transformation; must be invertible ($\det J \neq 0$) for a valid change of coordinates.
Einstein summation convention	爱因斯坦求和约定	An implied sum over any repeated index in a term, e.g. $\dot{x}^A = (\partial x^A / \partial q_i)\dot{q}_i$.
Lagrange multipliers	拉格朗日乘子	Auxiliary dynamical functions $\lambda^\alpha(t)$ introduced to enforce constraints without eliminating coordinates.
Extended Lagrangian	扩展拉格朗日量	$\mathcal{L}' = \mathcal{L} + \lambda^\alpha f_\alpha$; treating λ^α as coordinates makes the constraint $f_\alpha = 0$ an equation of motion.
Constraint force	约束力	The term $\lambda^\alpha \, \partial f_\alpha / \partial x^A$ appearing on the right-hand side of the extended Euler–Lagrange equation.
Reduced Lagrangian	约化拉格朗日量	The Lagrangian expressed only in the independent generalized coordinates after substituting the constraints; gives the same equations of motion as the extended formalism.
Kronecker delta	克罗内克 δ	$\delta_{ij} = 1$ if $i = j$ and 0 otherwise; used to simplify expressions involving coordinate derivatives.
Generalized momentum	广义动量	$p_i = \partial L / \partial \dot{q}_i$; the Lagrangian analogue of momentum conjugate to the coordinate q_i.
Generalized force	广义力	$\partial L / \partial q_i$; plays the role of force in the Euler–Lagrange equation $\dot{p}_i = \partial L / \partial q_i$.
Cyclic coordinate	循环坐标	A coordinate q_i that does not appear explicitly in L ($\partial L / \partial q_i = 0$); also called an ignorable coordinate.
Ignorable	可忽略坐	Same as cyclic coordinate; its conjugate momentum is conserved, so

coordinate	标	it can be partly ignored when solving the equations.
Conservation law	守恒定律	A statement that a dynamical quantity remains constant along the physical motion, e.g. $p_i = \text{constant}$ when q_i is cyclic.
Noether's theorem	诺特定理	Every continuous symmetry of the Lagrangian gives rise to a conserved quantity; spatial invariance yields momentum conservation, time invariance yields energy conservation.
Hamiltonian	哈密顿量	Defined by $H = \sum_i p_i \dot{q}_i - L$; a function of coordinates and momenta obtained from the Lagrangian by a Legendre transform.
Legendre transform	勒让德变换	The mathematical operation that changes the independent variables from $(q, \dot{q})$ to (q, p), exchanging velocities for momenta.
Conservation of the Hamiltonian	哈密顿量的守恒	If $\partial L/\partial t = 0$, then $dH/dt = 0$ and H is a constant of the motion.
Euler's theorem for homogeneous functions	欧拉齐次函数定理	If f is homogeneous of degree k, then $\sum_i x_i \left(\partial f/\partial x_i\right) = kf$; gives $\sum_i \dot{q}_i \left(\partial T/\partial \dot{q}_i\right) = 2T$ for natural coordinates.
Effective potential energy	有效势能	$U_{\text{eff}}(\rho)$; combines the real potential with centrifugal-like terms to reduce the problem to one-dimensional energy balance.
Stable equilibrium	稳定平衡	A configuration where the effective potential has a local minimum; small displacements produce restoring motion.
Unstable equilibrium	不稳定平衡	A configuration where the effective potential has a local maximum; small displacements cause the system to move away.
Neutral equilibrium	随遇平衡	A configuration where the effective potential is flat; the system can stay at rest at any nearby point.
Critical angular velocity	临界角速度	The value $\omega_{\text{crit}} = \sqrt{2g\alpha}$ at which the stability of the bead on a rotating parabolic wire changes character.